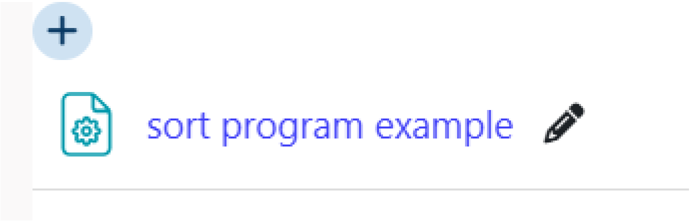
C Programming 01
Structure

Sample code for last week's sort is now available.
However, bubble sort and heap sort are not included.
Also, the code is combined into one, not separate.



Selection sorting and heap sorting will be one of the content questions in Report 2.

pre-learning

- If you haven’t watched the pre-learning videos yet, watch the videos below first.
  - https://wsdmoodle.waseda.jp/mod/millvi/view.php?id=5822907

Exercise 1

Please define the structure so that this program will work.
The specification of each member is as follows.
id: Unique ID of the book, integer, value must be greater than or equal to 0
name: Name of the book, characters (up to 20)
num: the number of books in stock, integer, value must be greater than or equal to 0

```
#include <stdio.h>
/* Structure Declaration */

int main(void)
{
        int i;
        struct  books books1[5];     /* Declaring a Structure Array */
        /* Enter data in structure array with scanf() */
        for(i = 0; i < 5; i++) {
                scanf("%d", &books1[i].id);
                scanf("%s", books1[i].name);
                scanf("%d", &books1[i].num);
        }
        for(i = 0; i < 5; i++) {
                printf("id:%d name:%s num:%d¥n", books1[i].id, books1[i].name, books1[i].num);
        }
        return 0;
}
```

Exercise 2

- Define a structure whose members are numerical values (1-12) representing what month it is and the average temperature (double) and create a program to input and display the month and temperature for four months.

Reference: Japan's average temperature (2023)

1	4.9	7	27.4
2	5.2	8	27.5
3	10.9	9	24.4
4	15.3	10	17.2
5	18.8	11	14.5
6	23.0	12	7.5

Exercise 3

Create a program with the following contents.

- Define a structure with the following members to represent a student's exam results:
  - Name (char type array)
  - Exam score (int type)
  - Grading (char type)
- Define a function to set the grading。
  - Take a pointer to the created structure as an argument.
  - Set letters in the grading based on the exam score.
  - 'A' → 80 points or more, 'B' → 70 points or more, 'C' → 60 points or more, 'D' → less than 60 points
- Enter the name and test score in the variables of the created structure.
- Set the grading by the function you created.
- View the contents of the structure (name, exam scores, grading).

Exercise4 (after Exercise3 is completed)

Using the structures and functions created in the previous question, create a program to enter the names and scores of four students, set the ratings, and display them.

Data for 4 students may be given by keyboard or array

About submitting Exercises

- Submission is voluntary.
- It is necessary to summarize some programs and execution results in Report 2.

- An example of today's Exercise program is shown at the beginning of next week's lecture.
- Next week's topic will be the Search algorithm.